



Capstone Project – SQL Task

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Introduction

This capstone is a milestone in your learning so far. In this project, you will apply SQL to analyse real-world data and extract meaningful insights.

SQL is the backbone of data-driven decision-making, and mastering it is essential for working with structured data efficiently. Employers expect developers and analysts to be comfortable writing complex queries, optimising performance, and transforming raw data into useful information.

You will be working with **Spotify streaming history data**, which captures user listening behaviour, track details, and playback patterns.

- **Dataset: Spotify Streaming History**

This is a chance to refine your skills and build a project that can be part of your portfolio. Take the time to experiment, think critically, and make your queries as insightful and efficient as possible!



Practical task

Spotify has hired you as a SQL analyst to improve user experience. They've given you raw playback data and want you to uncover insights that could help shape their algorithm.

Dataset download

- Please find the dataset here for your investigations: **Spotify Streaming History**.
- Once downloaded, move the file to your task folder so it is in your primary directory. This will make it easier to import without needing complicated file paths.

Setup Instructions for MariaDB Database Environment

Before working with queries, ensure your database environment is ready. These steps will guide you through setting up MariaDB, creating a database, importing data, and validating it. The **additional reading** provides detailed guidance on installing MariaDB and importing files. Please review that material before continuing here!

- **Database setup**

- Ensure **MariaDB** is installed and running on your system.
- Create a database and select it for use.

- **Tables and data import**

- Ensure a table exists for your playback data with the correct columns. If it doesn't exist, create one to match your dataset. If it does, confirm that its structure fits your data.
- Use the file import method outlined in the **Additional Reading** to load your CSV or data file into the playback data table. Confirm that all required fields are present in the imported data.

- **Data validation**

- Test the data with a simple query to ensure it's loaded correctly and check that the output matches your expectations (all columns should display properly).
- Look for missing values where data is expected, verify that data types are correct (e.g., numbers aren't stored as text), and ensure indexing is set up for efficiency.

Queries to run

1. **Skipped songs analysis**

- Find tracks that were skipped (skipped = 'TRUE') immediately after being played for less than 10 seconds, and count how many times this happened for each track.
- If a song has a lot of skips but a high total plays, what does that mean?

2. **Understanding user selections**

- Identify tracks that users manually selected the most (vs autoplay).
- Find songs that auto-played but had the highest skip rates—what's going wrong?

3. **Balance a playlist**

- Find artists that appear more than twice in your music data (overplay problem).
- What are the longest-played tracks that were played with shuffle disabled?

4. **Rage skip detector**

- Find sessions where a user skipped 5+ songs in under a minute (bad session).
- Is there a pattern? Are these specific artists, albums, or platforms?

5. **Artists with a single standout hit**

- Identify artists where 90%+ of plays come from only ONE track.
- Can you spot trends? Do people only listen to their biggest hit?

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.

For more information on saving queries, storing results, and preparing files for submission, refer to the **Additional Reading** file.



Share your thoughts

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